Bounded Perturbation Invariance for Deterministic Argmax

Research Architecture Runtime
`feature/lean-gated-operator-papers`

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Abstract

We study the finite deterministic argmax operator $\arg\max_{i\in[m]} s_i$ under additive score perturbations bounded in the ℓ_∞ norm. When a unique maximizer $i^\star$ exists with margin $\gamma(s) = s_{i^\star} - \max_{j\neq i^\star} s_j$, we prove that $\gamma(s) > 2\varepsilon$ is necessary and sufficient for $i^\star$ to remain the unique maximizer under every perturbation δ with $\|\delta\|_\infty \leq \varepsilon$. Repository publication is gated on Lean kernel check (`LEAN_FULL`) of the Mathlib $\mathbb{R}$ core (`REAL_MATHLIB`) in `Research.Operators.Argmax.Margin` (`margin_invariance`, `margin_sharpness`).

1 Introduction

Selection operators such as argmax are unstable near ties: infinitesimal score noise can change the reported index. A quantitative margin condition restores invariance under bounded noise. This note records the classical ℓ_∞ margin theorem as the first operator-specific result exercised end-to-end through the Discovery and Verification architectures in this repository.

2 Problem formulation

Fix $m \geq 2$ and scores $s \in \mathbb{R}^m$. Write $[m] = \{1, \ldots, m\}$. The (set-valued) argmax is

$$\operatorname{Argmax}(s) = \Big\{ i \in [m] : s_i = \max_{j\in[m]} s_j \Big\}.$$

We write $i^\star = \arg\max_i s_i$ when $|\operatorname{Argmax}(s)| = 1$.

Perturbation model. For $\varepsilon \geq 0$, admissible perturbations satisfy $\|\delta\|_\infty = \max_i |\delta_i| \leq \varepsilon$. The perturbed scores are $s + \delta$.

3 Notation and definitions

Definition 1 (Margin)**.** *If $i^\star$ is the unique maximizer of s, the margin is*

$$\gamma(s) = s_{i^\star} - \max_{j\neq i^\star} s_j.$$

Otherwise $\gamma(s)$ is undefined.

4 Mathematical assumptions

1. $m \geq 2$ and $s \in \mathbb{R}^m$.
2. A unique maximizer $i^\star$ exists, so $\gamma(s) > 0$.
3. Perturbations obey $\|\delta\|_\infty \leq \varepsilon$ with $\varepsilon \geq 0$.

We do not treat data-dependent scores $s_i(D)$, other ℓ_p balls, or set-valued selection rules beyond recording that ties exclude the hypothesis.

5 Main theorem

Theorem 2 (Bounded perturbation invariance)**.** *Under the assumptions above, if $\gamma(s) > 2\varepsilon$, then for every admissible δ, $i^\star$ is the unique maximizer of $s + \delta$.*

Theorem 3 (Sharpness)**.** *Under the same assumptions, if $\gamma(s) \leq 2\varepsilon$, then there exists an admissible δ such that $i^\star$ is not the unique maximizer of $s + \delta$.*

6 Supporting lemmas

Lemma 4 (Worst-case gap)**.** *For any admissible δ and any $j \neq i^\star$,*

$$(s + \delta)_{i^\star} - (s + \delta)_j \geq s_{i^\star} - s_j - 2\varepsilon \geq \gamma(s) - 2\varepsilon.$$

Proof. $(s + \delta)_{i^\star} \geq s_{i^\star} - \varepsilon$ and $(s + \delta)_j \leq s_j + \varepsilon$. Subtracting yields the first inequality; the second uses the definition of γ. □

7 Proofs

Proof of Theorem 2. Lemma 4 and $\gamma(s) > 2\varepsilon$ imply $(s + \delta)_{i^\star} > (s + \delta)_j$ for all $j \neq i^\star$. □

Proof of Theorem 3. Let $j^\star \in \operatorname{Argmax}_{j \neq i^\star} s_j$. Set $\delta_{i^\star} = -\varepsilon$, $\delta_{j^\star} = +\varepsilon$, and $\delta_k = 0$ otherwise. Then $\|\delta\|_\infty \leq \varepsilon$ and

$$(s + \delta)_{i^\star} - (s + \delta)_{j^\star} = \gamma(s) - 2\varepsilon \leq 0,$$

so $i^\star$ is not a unique maximizer. □

8 Discussion

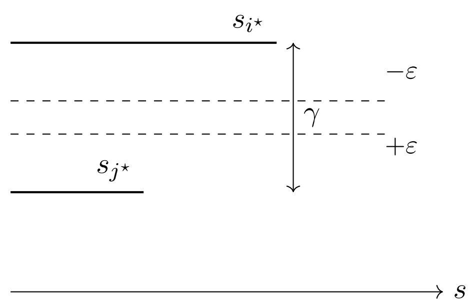
8.1 Intuition

The margin is a buffer against adversarial noise that simultaneously depresses the winner and elevates a runner-up. Each direction can consume at most ε, hence the factor two.

8.2 Geometric explanation

8.3 Why the theorem matters

It gives an exact, checkable certificate that a reported argmax is stable to bounded score noise—useful for post-hoc selection and audit of ranking pipelines.



Invariance requires $\gamma > 2\varepsilon$

Figure 1: Margin buffer versus ℓ_∞ adversarial moves.

8.4 Where assumptions are used

Uniqueness supplies γ; ℓ_∞ yields the additive 2ε shrinkage; finiteness of m is used only to ensure maxima exist. Indices in the paper are $[m] = \{1, \ldots, m\}$; the reference implementation uses $0..m-1$ with the obvious shift.

8.5 Examples

Take $s = (3, 1, 0.5)$ so $\gamma = 2$. For $\varepsilon = 0.9$, invariance holds. For $\varepsilon = 1$, $\delta = (-1, +1, 0)$ yields $s + \delta = (2, 2, 0.5)$ (tie).

8.6 Counterexamples and boundary cases

At $\gamma = 2\varepsilon$ the adversary forces a tie: uniqueness fails though $i^\star$ may remain *a* maximizer. Ties in s exclude the theorem hypothesis entirely.

9 Interpretation

The condition $\gamma > 2\varepsilon$ is both a proof target and an operational test: compute the empirical margin and compare to the noise budget.

10 Utility implications

Stabilization mechanisms (score smoothing, hysteresis, top-k reporting) can be justified when the measured margin is too small relative to ε.

11 Limitations

- Lean formalization is a Mathlib $\mathbb{R}$ core (`REAL_MATHLIB`) scoped to the margin theorems.
- Finite score vectors; abstract s, not $s_i(D)$.
- ℓ_∞ only; other norms change constants.
- Engineering float checks can blur exact equality $\gamma = 2\varepsilon$; Lean uses exact $\mathbb{R}$.

12 Open questions

Extensions to ties (set-valued argmax), ℓ_p balls, and data-dependent scores $s_i(D)$; composition with top-k and thresholding.

13 Connections to existing framework

Artifacts are minted as typed Discovery IR, packaged as a sealed CRP, System B discharges obligations, then System 3 runs `lake build` + axiom capture. Lean modules: `Research.Operators.Argmax.Margin`. Certificate: `lean/certificates/argmax/bounded-perturbation-margin/`.

14 Verifier summary

Publication requires `derived_lean_status=LEAN_FULL` (sorry/admit = 0, `build_ok`, axiom closure captured). Kernel axioms observed: `propext`, `Quot.sound`. Prior computational-only paper archived under `_archive/v0-computational/`.

15 Appendix

Canonical statement (repository string):

> Let $m \geq 2$, $s \in \mathbb{R}^m$ with unique maximizer $i^\star = \arg\max_i s_i$ and margin $\gamma(s) = s_{i^\star} - \max_{j \neq i^\star} s_j > 0$. Let $\varepsilon \geq 0$ and $\delta \in \mathbb{R}^m$ with $\|\delta\|_\infty \leq \varepsilon$. If $\gamma(s) > 2\varepsilon$, then $i^\star$ is the unique maximizer of $s + \delta$.

References

References

[1] Repository implementation: `implementation/src/operators/argmax/` and `implementation/docs/operators/argmax.md`.